



SAL-O-MAK

Broad Spectrum Feed and Gut Acidifer



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Sal-O-Mak is a perfect blend of buffered organic acid mixture, specialized for poultry & aqua, enhanced with essential oils & esterified coated butyric acid. This addition makes the product effective in the finished feed as well as whole intestinal tract. This contributes to the birds stable feed digestion while allowing important digestive enzymes to be activated faster helps in promoting bird's production and growth.

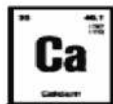
Poultry Feed always contains higher protein values. Protein has buffer effect which absorbs the acids from intestine, which results in higher pH of intestine. With ideal composition of organic acids and advanced blending technology, Sal-O-Mak remarkably reduce the pH of intestine and improves efficacy of enzymes by enhancing release and utilization of nutrients from feed.

COMPOSITION

Salts of Citric acid, Propionic acid, Esterified Coated butyric acid, Formic acid, Acetic acid, Fumaric acid, Benzoic acid & Essential oils

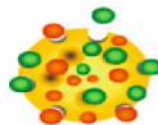
Next generation acidifiers

Combination of 3 (or more) organic acids

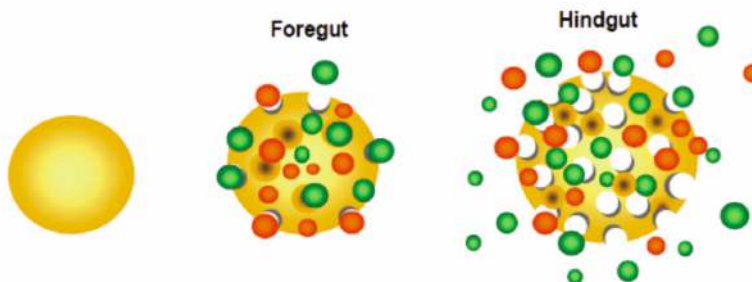


Calcium, Potassium and Sodium salts of Organic Acids have Lower PKa Value

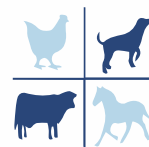
Micro-encapsulation with essential oils



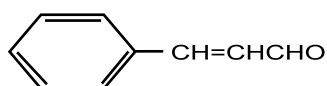
Release of micro-encapsulated organic acids and essential oils



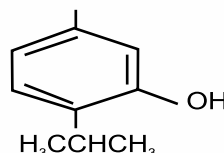
Fat capsule digestion through Lipase



Essential oils are secondary plant metabolites. EO's enhance production of digestive secretions, stimulate blood circulation, exert antioxidant properties, reduce levels of pathogenic bacteria by increase permeability of the cell wall of pathogenic bacteria and enhance immune status.



Antimicrobial
Antiinflammatory
Antispasmodic
Antiurease
Cancer-preventive
Flavor
Hypoglycemic
Cinnamaldehyde



Antimicrobial
Antiinflammatory
Antimelanomic
Antioxidant
Antiseptic
Carminative
Flavor
Thymol

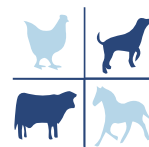
BENEFITS

- **To Control: *E. Coli*, *Salmonella*, *Clostridia* etc.**
Sal-O-Mak's non-dissociated organic acids can penetrate the bacterial cell wall and destroy the normal physiological functions of pH sensitive bacteria.
- **Improve Nutrient Digestibility:**
Digestibility and availability of nutrients (such as Ca and P) due to developing beneficial microflora (*Lactobacillus* spp.)
- **Esterified Coated Butyric Acid in Sal-O-Mak:**
In coated form enables Sal-O-Mak to act effectively in hind gut. Increases the villus width, height and area of the duodenum, jejunum and ileum of broiler chicks.
- Some acids are in liquid form and highly pungent & irritant acids. Sal-o-mak has it, in **sustained release micro pellets** makes it less pungent & easy for handling in feed mills
- Stimulation & activation of digestive enzymes by lowering gut pH & induces the release of digestive enzymes
- Better villi development & aids in absorption of nutrients which leads to appreciable improvement in production parameters like FCR & body weight

ADVANTAGES

- Sal-O-Mak is a Proprietary blend of buffered **organic acid** with **Essential oils** and **Esterified Butyric acid**.
- Micro-Encapsulated Acidifier.
- Does not damage other ingredients in feed.
- Sal-O-Mak is highly diffusive and less corrosive for machineries due to its patented manufacturing technology.
- **Sal-O-Mak's acid concentration is 48-50%**
- Due to its synergistic action of Acidification with Essential Oil, its effective against *Salmonella*, *E.Coli*, *Clostridium* spp., *Pasteurella* etc..





Chemical properties of selected acids and salts

Organic acid	Salts	pk Value	Solubility	Molecular weight (/mol)	Gross energy (MJ/ kg)
Formic acid	Calcium formate, Potassium formate, Sodium formate	3.75	Very good	48.0	5.8
Acetic acid	Sodium acetate	4.76	Very good	60.1	14.8
Propionic acid	Calcium propionate	4.88	Very good	74.1	20.8
Lactic acid	Calcium lactate	3.90	good	90.1	15.1
Fumaric acid	Magnesium fumarate	3.04/4.44	low	116.1	11.5
Citric acid	Calcium citrate, zinc citrate, Sodium citrate	3.14/4.76/6.40	good	210.1	10.3

HOW DO SAL-O-MAK WORKS?

The antibacterial activity of Sal-O-Mak is related to the reduction of pH, as well as their ability to penetrate pathogenic cell wall. In the dissociated form organic acids are lipids soluble in which they are able to enter microbial cell. Once an acid enter into the cell wall releases the proton in more alkaline environment, reduces the intercellular pH. This influences microbial metabolism, leading to an intercellular accumulation of acids actions. The acid anion seems to be very important for the antibacterial effect.

The antibacterial effect of dietary organic acids in chicken is mainly take place in crop & gizzard. Where as Sal-O-Mak blend with esterified butyric acid along with essential oils act also well in small and large intestine of GIT.

APPLICATION

- Feed Preservation as mould inhibitor.
- Gut PH Reduction as gut acidifier.
- Better villi development and aids in absorption of nutrients.
- Pathogenic bacterial control.

DOSAGE

- Broiler / Layers : 500 g - 1kg / ton of feed
- Breeders : 1kg - 2kg / ton of feed

PRESENTATION

25kg bag.

SHELF LIFE

24 Months from Date of Manufacture



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

Manufactured and Marketed By :

ZENMAK NUTRIGENCIES AND HEALTH PVT. LTD.

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